In [1]:
```
!wget -q https://data.vision.ee.ethz.ch/cvl/DIV2K/DIV2K_valid_HR.zip
!unzip -q DIV2K_valid_HR.zip
```

In [2]:
```
!python srcnn.py train --data_dir DIV2K_valid_HR --out_dir results --epochs
100
```

```
Device: cuda
Split: train=60 val=20 test=20
epoch   1 train_mse 0.17606 val_mse 0.02680 val_psnr 15.93 dB
epoch   5 train_mse 0.02462 val_mse 0.02775 val_psnr 15.72 dB
epoch  10 train_mse 0.01766 val_mse 0.01591 val_psnr 18.33 dB
epoch  15 train_mse 0.01252 val_mse 0.01158 val_psnr 19.72 dB
epoch  20 train_mse 0.00940 val_mse 0.00869 val_psnr 21.01 dB
epoch  25 train_mse 0.00735 val_mse 0.00700 val_psnr 21.90 dB
epoch  30 train_mse 0.00641 val_mse 0.00617 val_psnr 22.46 dB
epoch  35 train_mse 0.00600 val_mse 0.00608 val_psnr 22.50 dB
epoch  40 train_mse 0.00594 val_mse 0.00612 val_psnr 22.39 dB
epoch  45 train_mse 0.00523 val_mse 0.00498 val_psnr 23.48 dB
epoch  50 train_mse 0.00480 val_mse 0.00460 val_psnr 23.87 dB
epoch  55 train_mse 0.00447 val_mse 0.00428 val_psnr 24.22 dB
epoch  60 train_mse 0.00531 val_mse 0.00652 val_psnr 22.01 dB
epoch  65 train_mse 0.00424 val_mse 0.00397 val_psnr 24.58 dB
epoch  70 train_mse 0.00389 val_mse 0.00376 val_psnr 24.85 dB
epoch  75 train_mse 0.00372 val_mse 0.00362 val_psnr 25.02 dB
epoch  80 train_mse 0.00370 val_mse 0.00354 val_psnr 25.11 dB
epoch  85 train_mse 0.00346 val_mse 0.00338 val_psnr 25.37 dB
epoch  90 train_mse 0.00333 val_mse 0.00326 val_psnr 25.57 dB
epoch  95 train_mse 0.00325 val_mse 0.00318 val_psnr 25.68 dB
epoch 100 train_mse 0.00321 val_mse 0.00319 val_psnr 25.63 dB
{
  "seed": 42,
  "lr": 0.001,
  "batch_size": 16,
  "epochs": 100,
  "best_val_psnr": 25.76216697692871,
  "test_psnr_srcnn": 26.867746353149414,
  "test_mse_srcnn": 0.0024385303258895874,
  "test_psnr_nearest": 26.17022705078125,
  "test_mse_nearest": 0.0028156007174402475,
  "test_psnr_bilinear": 26.761913299560547,
  "test_mse_bilinear": 0.002527223201468587,
  "test_psnr_bicubic": 27.8586483001709,
  "test_mse_bicubic": 0.001989213516935706
}
Saved results.json, curves.png, samples.png, best_srcnn.pt to results
```

In [3]:
```
!python srcnn.py demo --image DIV2K_valid_HR/0801.png --ckpt
results/best_srcnn.pt
```

```
Bicubic PSNR: 29.11 dB | SRCNN PSNR: 28.45 dB
Saved results/demo_output.png
```

In [4]:
```
!python srcnn.py train --data_dir DIV2K_valid_HR --out_dir results_bs2 --
epochs 100 --batch_size 2
```

```
Device: cuda
Split: train=60 val=20 test=20
epoch   1 train_mse 0.04510 val_mse 0.02054 val_psnr 17.17 dB
epoch   5 train_mse 0.00784 val_mse 0.00673 val_psnr 22.10 dB
epoch  10 train_mse 0.00467 val_mse 0.00442 val_psnr 24.01 dB
epoch  15 train_mse 0.00374 val_mse 0.00363 val_psnr 24.98 dB
epoch  20 train_mse 0.00439 val_mse 0.00376 val_psnr 24.76 dB
epoch  25 train_mse 0.00318 val_mse 0.00310 val_psnr 25.79 dB
epoch  30 train_mse 0.00293 val_mse 0.00287 val_psnr 26.15 dB
epoch  35 train_mse 0.00286 val_mse 0.00284 val_psnr 26.09 dB
epoch  40 train_mse 0.00272 val_mse 0.00266 val_psnr 26.52 dB
epoch  45 train_mse 0.00267 val_mse 0.00262 val_psnr 26.59 dB
epoch  50 train_mse 0.00257 val_mse 0.00252 val_psnr 26.77 dB
epoch  55 train_mse 0.00256 val_mse 0.00247 val_psnr 26.91 dB
epoch  60 train_mse 0.00245 val_mse 0.00246 val_psnr 26.85 dB
epoch  65 train_mse 0.00274 val_mse 0.00251 val_psnr 26.69 dB
epoch  70 train_mse 0.00293 val_mse 0.00262 val_psnr 26.46 dB
epoch  75 train_mse 0.00254 val_mse 0.00241 val_psnr 26.99 dB
epoch  80 train_mse 0.00250 val_mse 0.00241 val_psnr 26.95 dB
epoch  85 train_mse 0.00236 val_mse 0.00232 val_psnr 27.18 dB
epoch  90 train_mse 0.00235 val_mse 0.00232 val_psnr 27.19 dB
epoch  95 train_mse 0.00231 val_mse 0.00229 val_psnr 27.24 dB
epoch 100 train_mse 0.00231 val_mse 0.00226 val_psnr 27.33 dB
{
  "seed": 42,
  "lr": 0.001,
  "batch_size": 2,
  "epochs": 100,
  "best_val_psnr": 27.3289852142334,
  "test_psnr_srcnn": 28.483928680419922,
  "test_mse_srcnn": 0.0017114225775003433,
  "test_psnr_nearest": 26.17022705078125,
  "test_mse_nearest": 0.0028156007174402475,
  "test_psnr_bilinear": 26.761913299560547,
  "test_mse_bilinear": 0.002527223201468587,
  "test_psnr_bicubic": 27.8586483001709,
  "test_mse_bicubic": 0.001989213516935706
}
Saved results.json, curves.png, samples.png, best_srcnn.pt to results_bs2
```

In [5]:
```
!python srcnn.py demo --image DIV2K_valid_HR/0801.png --ckpt
results_bs2/best_srcnn.pt --out_dir results_bs2
```

```
Bicubic PSNR: 29.11 dB | SRCNN PSNR: 30.27 dB
Saved results_bs2/demo_output.png
```